WAIRAKA MODERN SECONDARY SCHOOL-JINJA

S.4 END OF TERM 1, 2026

Uganda Certificate of Education

Physics 1 Theory

Time: 2Hours 30minutes

INSTRUCTIONS

This paper has **two** *sections;* **A** *and* **B** *It has* ***SIX*** *examination items.*

Section **A** *has* **TWO** *compulsory items.*

Section **B** *has* **two** *parts;* **I** *and* **II**. *Answer* **one** *item from* **each** *part.*

Answer **four** *items in all.*

Any additional item(s) answered will not be scored.

All answers **must** *be written in the booklets provided*

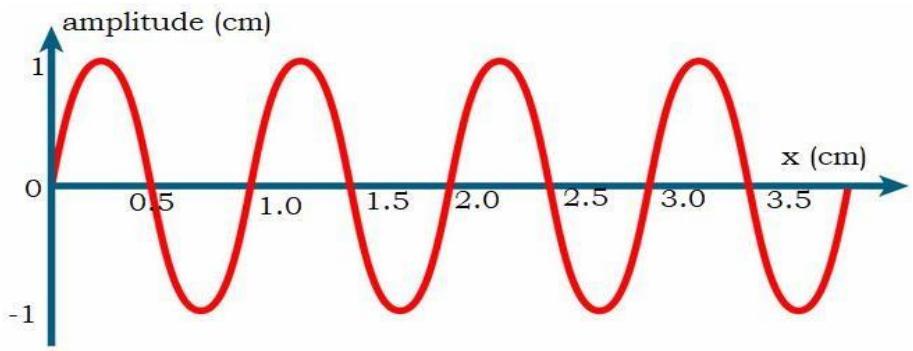
SECTION A

Attempt all items in this section

ITEM ONE

Learners of S.4 visited Kasenyi landing site and made critical observations which seemed unusual to them without valid scientific reasons. They observed the following;

- A magical pool of water along the way to the landing site on a tarmac road.
- The fish in the water appeared nearer and bigger than when outside.
- A stand-by computerized system housed on the lake shores
- A radar that emits radio waves was being used



The figure above shows a wave form from a computerized system

Hint:

- The frequency of the computerized system was, f = 180 Hz, and fishing would only continue if the velocity of the water waves in the lake was less than 7.2 kmh^{-1}.

Task: As a learner of physics; assume that the learners have approached you for help with the above information;

(a) Explain to the learners to understand

i) The occurrence of Magical pool of water along the way
ii) Why the fish in water appeared nearer than when outside?
iii) Why a stand-by computerized system was housed on the lake shores
iv) why A radar that emits radio waves was being used

(b) Establish whether fishing was possible on that day.

(c) Explain the properties and applications of the electromagnetic waves used by the radar.

ITEM TWO

While on a one-month trip to the coast in another country, a tourist was able to observe the following

- At night, the sky was dark and filled with many stars of different colors and brightness. He wondered how they came into existence and why they varied from each other and how they are important for human life
- The moon appeared almost every day but kept varying in appearance.
- The area he was in a wet season while his home area had been in a dry season for about three months. This made him wonder.
- The water levels kept increasing in level but reduced after about two hours. The tourist wanted to know if it was possible to make a telephone call to his family and he has also approached you to explain to him about his other issues.

Task

Use your knowledge of Physics to

(a) Help the tourist understand the observations he made.

(b) Help the tourist understand how it is possible to make telephone calls to his home country.

(c) Help the man identify ways in which the stars are useful to human life.

SECTION B

PART I

Attempt only <u>one</u> item from this part

ITEM THREE

A number of issues have sprung up at a construction site following the laying of a concrete slab

- One of the supervisors wearing sharp pointed shoes with an area of $\mathbf{40\ cm^2}$, found it hard to walk on a newly laid wet surface as compared to other wearing fiat shoes of $\mathbf{60cm^2}$

- One of the builders proposed that the concrete slab on the first floor needed to be reinforced unlike the one on the ground floor. The owner of the site did not agree with him but the builder had no satisfactory explanation on his proposal.

- Lifting of concrete blocks to use on the first floor has become hard and tedious. The workers claim the distance of $\mathbf{8m}$ is very high for them to throw the blocks. Some of them who attempted ended up breaking the concrete blocks as they hit the ground with very loud sound.

  In an effort to respond to one of the complaints, an engineer suggested a pulley system of four wheels be connected to a platform on which two blocks each of $\mathbf{8kg}$ could be put. The engineer was worried on the efficiency of the pulley system being less than $\mathbf{100}\%$ and how much effort will be required to lift the blocks

Hint

- ***Both supervisors have a mass of 70kg***
- ***Acceleration due to gravity is 10ms^{-2}***
- ***20% of the work input is wasted in the pulley system***

Use your knowledge of Physics to;

(a) Explain to the supervisor why it was harder for one of them to walk on the newly laid wet concrete surface.

(b) To help the home owner understand the builder's proposal of reinforcing the concrete on the first floor.

(c) Explain to the builders how to use the pulley system, and improve its efficiency

(d) Help the workers determine the minimum effort required to lift the blocks.

ITEM FOUR

Two Senegalese tourists went mountain climbing in Uganda. They were warned before climbing that if they climbed higher than 43,000 meters, they might experience nose bleeding. At sea level, the barometric reading is 76 cmHg, while at the top of the mountain, it is 72.12 cmHg. They made tea and cooked some food when they got to the top, but they had trouble getting the food cooked. Given that 294kJ of energy was supplied to boil 1litre of water initially at
26°C, the Specific heat capacity of water is 4200J/kg/K, Density of air is 1.2 kg m^{-3}.and the Density of mercury is 13600 kg m^{-3}.

Task:

As a physics student, help the tourists to

(a) Know whether they will bleed at the peak of the mountain using the mathematical approach, and explain why?
(b) Determine the temperature at which the water boils at the top of the mountain and explain the implications of this on boiling.
(c) Design an instrument that could be used to measure the pressure of the atmosphere to be used by the climber to avoid the eventualities as they approached the top of the mountain. (Available apparatus: bowel, a one-meter-long glass tube, and mercury)

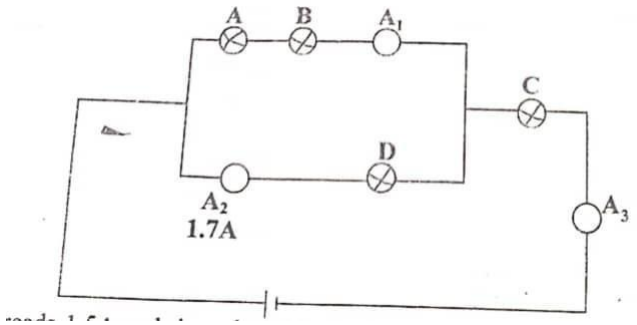
PART II

Attempt only <u>one</u> item from this part

ITEM FIVE

A head teacher of a certain secondary school constructed a residential house. The painting was done, electricity installation and water harvesting tanks put in place. The electrician connected different bulbs A, B, C and D in a circuit as shown below. Also, the head teacher does not know the effective resistance if **3Ω**, **4Ω**, and **5 Ω** are connected in parallel.

If ammeter $\mathbf{A_1}$ reads **1.5A** and $\mathbf{A_2}$ reads **1.7A,** the head teacher does not know what Ammeter $\mathbf{A_3}$ will read.



Further, during construction, many materials were being used and the Engineer left un-used items in the compound including; copper, plastic objects, wood, Rubber, Aluminium, and glass. The head teacher thinks that left materials can be used for other purposes but does not know whether they are electric conductors or insulators. He was also surprised when told that his charger uses DC voltage to charge his phone and yet the wall socket has 240 AC voltage.

Task:

Help the headteacher to:

a) Group the materials as insulators or conductors.
b) Obtain the ammeter reading $\mathbf{A_3}$
c) Obtain the effective resistance for resistors **3Ω, 4Ω** and **5Ω**
d) Understand how the voltage was changed from AC to DC voltage in the charger (include illustration)

ITEM 6

A car battery with an e.m.f. of 12.0V was procured and installed in a car to run the following components;

i. A car radio with components of total resistance 4.0 Ω
ii. Two head lights with a total resistance of 0.20 Ω
iii. Two hind lights of total resistance 0.35 Ω

After the components were run for some time, the battery ran down and it could not start the car engine.

Task:

As a learner of Physics;

(a) advise the car owner on the form of connection suitable for the above components (include illustrations and your working)
(b) guide the car owner, on what he can do such that the battery makes the car start again.
(c) suggest what the car owner can do to prolong the life span of his car battery.

NEVER LOOSE HOPE

THE END